

Tao 11

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1 Integration

Let $f : I \rightarrow \mathbb{R}$ be RI. Suppose $I = J \cup K$ is a partition of I into intervals. Then $f_1 = f|_J, f_2 = f|_K$ are RI and $\int_I f = \int_J f_1 + \int_K f_2$.

Proof: Suppose f^t is pc maj f . Then $f^t = f_1^t \oplus f_2^t$ for f_1^t, f_2^t pc maj f_1, f_2 . Then

$$pc \int_I f^t = pc \int_I f_1^t \oplus f_2^t = pc \int_J f_1^t + pc \int_K f_2^t \geq \overline{\int_J f_1} + \overline{\int_K f_2}.$$

Since this holds for all f^t , $\overline{\int_I f} \geq \overline{\int_J f_1} + \overline{\int_K f_2}$.

Conversely, suppose f_1^t, f_2^t are pc maj f_1, f_2 . Then $f^t = f_1^t \oplus f_2^t$ pc maj f .

$$pc \int_J f_1^t + pc \int_K f_2^t = pc \int_I f_1^t \oplus f_2^t = pc \int_I f^t \geq \overline{\int_I f}.$$

Since this holds for all f_1^t, f_2^t , we have

$$\overline{\int_J f_1} + \overline{\int_K f_2} \geq \overline{\int_I f}.$$

It follows $\overline{\int_I f} = \overline{\int_J f_1} + \overline{\int_K f_2}$. Similarly, $\underline{\int_I f} = \underline{\int_J f_1} + \underline{\int_K f_2}$. Since $\overline{\int_I f} = \underline{\int_I f}$ and upper Riemann sums are at least lower Riemann sums, f_1, f_2 are Riemann integrable.

If $f, g : I \rightarrow \mathbb{R}$ are RI, then so is $\max(f, g) : I \rightarrow \mathbb{R}$. Let $\epsilon > 0$. Pick pc f^t, f_b majmin f st $pc \int_I (f^t - f_b) < \epsilon/2$. Similarly, pick g^t, g_b st $pc \int_I (g^t - g_b) < \epsilon/2$. For all $x \in I$,

$$\max(f^t(x), g^t(x)) - \max(f_b(x), g_b(x)) \leq \max(f^t(x) - f_b(x), g^t(x) - g_b(x)),$$

Then

$$\begin{aligned}
pc \int_I \max(f^t, g^t) - \max(f_b, g_b) &\leq pc \int_I \max(f^t - f_b, g^t - g_b) \\
&\leq pc \int_I (f^t - f_b) + (g^t - g_b) = pc \int_I (f^t - f_b) + pc \int_I (g^t - g_b) \\
&< \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}$$

Since $\max(f^t, g^t), \max(f_b, g_b)$ pc majmin $\max(f, g)$, we're done.

Suppose $f, g : I \rightarrow \mathbb{R}$ are RI. Then fg is RI.

Proof: Let $f = f_+ + f_-$, $g = g_+ + g_-$. Suppose we show if $f, g : I \rightarrow \mathbb{R}_{\geq 0}$ are RI, then fg is RI. Then by scaling, we show $f_+g_+, f_+g_-, f_-g_+, f_-g_-$ are all RI, so $f_+g_+ + f_+g_- + f_-g_+ + f_-g_- = (f_+ + f_-)(g_+ + g_-) = fg$ is RI.

Let f, g be bdd by $M > 0$. Let $\epsilon > 0$. Pick f^t, f_b pc majmin f st $pc \int_I (f^t - f_b) < \epsilon/(2M)$ and $f^t, f_b \geq 0$. Pick similarly for g . Then $f^t g^t, f_b g_b$ pc majmin fg .

We have

$$\begin{aligned}
pc \int_I (f^t g^t - f_b g_b) &= pc \int_I (f^t g^t - f^t g_b + f^t g_b - f_b g_b) \\
&= pc \int_I f^t (g^t - g_b) + g_b (f^t - f_b) \leq pc \int_I M(g^t - g_b) + pc \int_I M(f^t - f_b) \\
&< \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}$$

Hence, fg is RI.

Let $f : I \rightarrow \mathbb{R}$ be uniformly continuous. Then f is RI.

Proof: First, I show f is bounded. By definition of uniform continuity, pick $\delta > 0$ such that for $x, y \in I$, $|x - y| < \delta \rightarrow |f(x) - f(y)| < 1$. Pick $x_1 < x_2 < \dots < x_n$ in I such that for all $x \in I$, there exists $i \in [n]$ such that $|x - x_i| < \delta$. The set $\{f(x_i)\}_{i=1}^n$ is finite, and therefore bounded, say by M .

For $x \in I$, pick $k \in [n]$ such that $|x - x_i| < \delta$. Then $|f(x)| \leq |f(x) - f(x_i)| + |f(x_i)| < 1 + M$, so f is bounded by $M + 1$.

Now I show f is RI. If $|I| = 0$, we're done. Else, $|I| > 0$. Let $\epsilon > 0$ and pick $\delta > 0$ such that $|x - y| < \delta \rightarrow |f(x) - f(y)| < \epsilon/(2|I|)$. Pick a partition P of I such that for all $J \in P$, $|J| < \delta$. Then we may pick pc f^t, f_b majmin f st $f^t - f_b \leq \epsilon/|I|$ over I .

Then

$$pc \int_I (f^t - f_b) \leq pc \int_I \epsilon/|I| = \epsilon$$

so f is RI.

Let $f : I \rightarrow \mathbb{R}$ be bdd cont. Then f is RI.

Proof: Suppose f is bdd by $M > 0$. I will show the result for $I = (a, b]$, $a < b$. Pick $0 < \epsilon < b - a$, so $f_1 = f|_{[a+\epsilon, b]}$ is RI. pc M maj f on $(a, a+\epsilon)$. Then $M \oplus f_1^t$ pc maj f on I , so

$$pc \int_I M \oplus f_1^t = pc \int_{(a, a+\epsilon)} M + pc \int_{[a+\epsilon, b]} f_1^t = M\epsilon + pc \int_{[a+\epsilon, b]} f_1^t \geq \overline{\int_I} f.$$

Take inf of both sides, so $M\epsilon + \int_{[a+\epsilon, b]} f_1 \geq \overline{\int_I} f$.

Similarly, $-M \oplus f_{1b}$ pc min f , so $-M\epsilon + \int_{[a+\epsilon, b]} f_1 \leq \underline{\int_I} f$. It follows $0 \leq \overline{\int_I} f - \underline{\int_I} f \leq 2M\epsilon$. Since ϵ can be taken arbitrarily small, $\overline{\int_I} f = \underline{\int_I} f$, ie f is RI.

Integral test. Let $f : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be monotone decreasing. Then $\sum_{n=0}^{\infty} f(n)$ converges iff $\sup_{N>0} \int_{[0, N]} f < \infty$.

Proof: Suppose $\sum_{n=0}^{\infty} f(n)$ converges. Then the sequence of parital sums $\{\sum_{n=0}^N f(n)\}$ is bdd. $\int_{[0, N]} f \leq \sum_{n=1}^N f(n) = -f(0) + \sum_{n=0}^N f(n)$, so $\{\int_{[0, N]} f\}$ is bdd.

Conversely, suppose $\sup\{\int_{[0, N]} f\} < \infty$. Then $\sum_{n=0}^N f(n) \leq \int_{[0, N+1]} f$ is bdd above.

Fundamental theorem of Calculus, part I. Let $a < b$, $I = [a, b]$. Let $f : I \rightarrow \mathbb{R}$ be RI. Define $F : I \rightarrow \mathbb{R}$ by $F(x) = \int_{[a, x]} f$ for all $x \in I$. Then F is cont and if f is cont at $x_0 \in I$, then F is diff at x_0 with $F'(x_0) = f(x_0)$.

Proof: Since f is RI on $[a, b]$, f is RI on $[a, x]$, so F is well defined. Let f be bdd by $M > 0$. WLOG $x \leq y$ in I . Then

$$F(y) - F(x) = \int_{[a, y]} f - \int_{[a, x]} f = \int_{(x, y]} f \leq \int_{(x, y]} M = M(y - x).$$

Similarly, $F(y) - F(x) \geq -M(y - x)$, so either way $|F(y) - F(x)| \leq M|y - x|$. Hence, F is Lipschitz continuous.

Suppose f is cont at x_0 . Let $\epsilon > 0$ and pick $\delta > 0$ st $x \in V_{\delta}(x_0) \cap I \rightarrow f(x) \in V_{\epsilon}(f(x_0))$. Let $x \in V_{\delta}(x_0) \cap (I - x_0)$. If $x > x_0$, then

$$F(x) - F(x_0) = \int_{(x_0, x]} f \leq \int_{(x_0, x]} f(x_0) + \epsilon = (x - x_0)(f(x_0) + \epsilon),$$

so $\frac{F(x) - F(x_0)}{x - x_0} \leq f(x_0) + \epsilon$. Similarly, $\frac{F(x) - F(x_0)}{x - x_0} \geq f(x_0) - \epsilon$, so $\frac{F(x) - F(x_0)}{x - x_0} \in V_{\epsilon}(f(x_0))$.

Now suppose $x < x_0$. Then

$$F(x_0) - F(x) = \int_{(x, x_0]} f \leq \int_{(x, x_0]} f(x_0) + \epsilon = (x_0 - x)(f(x_0) + \epsilon),$$

so $\frac{F(x_0) - F(x)}{x_0 - x} \leq f(x_0) + \epsilon$. Similarly, $\frac{F(x_0) - F(x)}{x - x_0} \geq f(x_0) - \epsilon$, so $\frac{F(x) - F(x_0)}{x - x_0} \in \overline{V_\epsilon(f(x_0))}$. Either way, $F'(x_0) = \lim_{x \rightarrow x_0; x \in I - x_0} \frac{F(x) - F(x_0)}{x - x_0} = f(x_0)$.